

Question 1

The beam AB shown in Figure Q1 is simply supported by a pin at A and a roller at B. The beam has a span of 4.0 m. It carries a vertical point load of 20 kN located 1.5 m from support A, and a uniformly distributed load (UDL) of 10 kN/m acting from 2.0 m to 4.0 m from A. The self-weight of the beam may be ignored.

- (a) Determine the vertical support reactions at A and B.
- (b) Draw the shear force diagram, showing all significant values.
- (c) Draw the bending moment diagram, stating the maximum bending moment and its position.
- (d) Sketch the deflected shape of the beam.

Figure Q1